

Proofs for Theoretical Guarantees: Belief-Conditioned Symmetries

Preliminaries and Assumptions

We formulate the Partially Observable Markov Decision Process (POMDP) as a fully observable Belief MDP with belief state space $\mathcal{B}$ and action space $\mathcal{A}$. Let $c_\theta : \mathcal{B} \rightarrow \mathcal{C}$ be the context extractor, and $\phi : \mathcal{C} \rightarrow G$ map the context to a group action. For a given belief state $b \in \mathcal{B}$, the inferred local gauge symmetry is denoted as $g_b = \phi(c_\theta(b)) \in G$.

We assume the group action on the belief space is measure-preserving, meaning $d(g_b \cdot b') = db'$. Furthermore, we make the following core assumptions regarding the contextual equivariance of the environment dynamics:

Assumption 1 (Belief Reward Invariance). *For any belief state $b \in \mathcal{B}$ and action $a \in \mathcal{A}$, the expected reward is invariant under the inferred local symmetry g_b :*

$$\bar{R}(b, a) = \bar{R}(g_b \cdot b, g_b \cdot a) \quad (1)$$

Assumption 2 (Belief Transition Equivariance). *The belief transition probability $P(b' | b, a)$ is equivariant under the inferred local symmetry g_b :*

$$P(b' | b, a) = P(g_b \cdot b' | g_b \cdot b, g_b \cdot a) \quad (2)$$

Proofs of Main Theorems

Theorem 1 (Contextual Value Invariance). *Under Assumptions 1 and 2, the optimal state-action value function $Q^*(b, a)$ and the optimal value function $V^*(b)$ are invariant under the inferred local gauge symmetry g_b :*

$$Q^*(b, a) = Q^*(g_b \cdot b, g_b \cdot a) \quad (3)$$

$$V^*(b) = V^*(g_b \cdot b) \quad (4)$$

Proof. We prove this using mathematical induction on the value iteration steps via the Bellman Optimality Operator $\mathcal{T}$.

Base Case ($k = 0$): Initialize the Q-function as zero for all states and actions, i.e., $Q_0(b, a) = 0, \forall b \in \mathcal{B}, a \in \mathcal{A}$. Trivially, $Q_0(b, a) = Q_0(g_b \cdot b, g_b \cdot a) = 0$ holds.

Inductive Step ($k \rightarrow k + 1$): Assume that at iteration k , the Q-function satisfies local invariance: $Q_k(b, a) = Q_k(g \cdot b, g \cdot a)$ for any $g \in G$. The Bellman update for iteration $k + 1$ is given by:

$$Q_{k+1}(b, a) = \bar{R}(b, a) + \gamma \int_{\mathcal{B}} P(b' | b, a) \max_{a' \in \mathcal{A}} Q_k(b', a') db' \quad (5)$$

Now, evaluate Q_{k+1} at the transformed state and action ($g_b \cdot b, g_b \cdot a$):

$$Q_{k+1}(g_b \cdot b, g_b \cdot a) = \bar{R}(g_b \cdot b, g_b \cdot a) + \gamma \int_{\mathcal{B}} P(b'' | g_b \cdot b, g_b \cdot a) \max_{a''} Q_k(b'', a'') db''$$

Applying Assumption 1, we have $\bar{R}(g_b \cdot b, g_b \cdot a) = \bar{R}(b, a)$. Next, we apply a change of variables in the integral. Let $b'' = g_b \cdot b'$. Since the group action is measure-preserving, $db'' = db'$. Substituting this into the integral yields:

$$Q_{k+1}(g_b \cdot b, g_b \cdot a) = \bar{R}(b, a) + \gamma \int_{\mathcal{B}} P(g_b \cdot b' | g_b \cdot b, g_b \cdot a) \max_{a''} Q_k(g_b \cdot b', a'') db'$$

By Assumption 2 (Transition Equivariance), $P(g_b \cdot b' | g_b \cdot b, g_b \cdot a) = P(b' | b, a)$. Since g_b is invertible and the action space is assumed to be closed under the group action, finding the maximum over a'' is equivalent to finding the maximum over a transformed action $a' = g_b^{-1} \cdot a''$, which implies $a'' = g_b \cdot a'$. Therefore:

$$\max_{a''} Q_k(g_b \cdot b', a'') = \max_{a'} Q_k(g_b \cdot b', g_b \cdot a')$$

Using the inductive hypothesis, $Q_k(g_b \cdot b', g_b \cdot a') = Q_k(b', a')$. Substituting all these simplifications back gives:

$$Q_{k+1}(g_b \cdot b, g_b \cdot a) = \bar{R}(b, a) + \gamma \int_{\mathcal{B}} P(b' | b, a) \max_{a'} Q_k(b', a') db'$$

Comparing this with Equation 5, we conclude that $Q_{k+1}(g_b \cdot b, g_b \cdot a) = Q_{k+1}(b, a)$.

Limit to Optimality: As $k \rightarrow \infty$, by the Contraction Mapping Theorem, $Q_k \rightarrow Q^*$. Thus, the optimal Q-function inherits the invariance: $Q^*(b, a) = Q^*(g_b \cdot b, g_b \cdot a)$. Consequently, $V^*(b) = \max_a Q^*(b, a) = \max_a Q^*(g_b \cdot b, g_b \cdot a) = V^*(g_b \cdot b)$. $\square$

Theorem 2 (Contextually Equivariant Policy). *Under Assumptions 1 and 2, the optimal deterministic policy π^* is conditionally equivariant to the inferred local gauge symmetry g_b :*

$$\pi^*(g_b \cdot b) = g_b \cdot \pi^*(b) \quad (6)$$

Proof. By definition, the optimal policy π^* greedily maximizes the optimal Q-function:

$$\pi^*(b) = \arg \max_{a \in \mathcal{A}} Q^*(b, a)$$

Evaluating the policy at the transformed belief state $g_b \cdot b$:

$$\pi^*(g_b \cdot b) = \arg \max_{\tilde{a} \in \mathcal{A}} Q^*(g_b \cdot b, \tilde{a})$$

Let $\tilde{a} = g_b \cdot a$. Since g_b is a bijection on the action space, maximizing over $\tilde{a}$ is equivalent to maximizing over a :

$$\pi^*(g_b \cdot b) = \arg \max_{g_b \cdot a} Q^*(g_b \cdot b, g_b \cdot a)$$

By Theorem 1, we know $Q^*(g_b \cdot b, g_b \cdot a) = Q^*(b, a)$. Substituting this in gives:

$$\pi^*(g_b \cdot b) = \arg \max_{g_b \cdot a} Q^*(b, a)$$

Since g_b acts as a linear transformation on the action space that is independent of the maximization objective $Q^*(b, a)$, we can extract it from the arg max operator:

$$\pi^*(g_b \cdot b) = g_b \cdot \left(\arg \max_{a \in \mathcal{A}} Q^*(b, a) \right) = g_b \cdot \pi^*(b)$$

This completes the proof. □